

The background features a light gray grid. Scattered across the grid are several 3D skeletal models of human figures in various poses, rendered in different colors: blue, green, pink, and cyan. On the right side, there is a white sphere with three arrows originating from it: a green arrow pointing upwards and to the right, a blue arrow pointing upwards and to the left, and a red arrow pointing downwards and to the left.

Progress Update – Capstone Project 2025/2026

Human Pose Estimation

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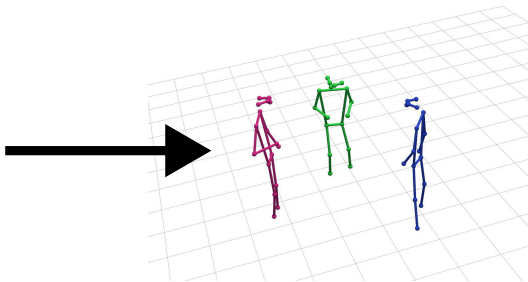
Task – Human Pose Estimation

- ▶ Given:

- Multiple video sequences containing multiple people.
- Recorded using multiple calibrated cameras.

- ▶ Reconstruct:

- 3D coordinates of skeletal joint for each subject.
- Movement of skeletal joint over time.



What to do next?

- ▶ **Summary:**

- Joint spatiotemporal modeling (ConvGRU) offers the best accuracy.
- However, stacked CNN may be preferred, as lower-resource alternative for real-time use.

- ▶ **Limitations:**

- Blurry predictions due to loss function.
- High computational cost for recurrent models.

- ▶ **Possible Future Improvements:**

- Use SSIM or GAN (Adversarial) loss to sharpen images.
- Incorporate raw satellite channels or weather data.
- Train natively with autoregressive feedback.
- Use more advanced recurrent units, e.g., TrajGRU or LSTA.

